

INTRO TO CRITICAL INFRASTRUCTURE FOR IT LEARNERS

LAB 04

Introduction to Supervisory Control and Data Acquisition (SCADA)

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1 Objective

The objective of this laboratory was to understand the basic architecture and operation of a SCADA system. The practical exercise involved connecting a FUXA HMI on the SCADA virtual machine to an OpenPLC Runtime on the PLC virtual machine using Modbus TCP.

The laboratory also involved loading a tank-fill simulation and observing the process through the SCADA/HMI interface.

2 Laboratory Environment

The laboratory was performed using the IC DFA-provided isolated virtual machines.

System	IP Address	Role
PLC VM	192.168.1.10	OpenPLC Runtime
SCADA VM	192.168.2.10	FUXA SCADA/HMI
Communication	Modbus TCP	PLC–SCADA communication

Table 1: Laboratory environment

3 PLC Configuration

The OpenPLC Runtime was accessed from the PLC virtual machine. The `TankFillSim.st` program was uploaded and compiled. The program was then started from the OpenPLC dashboard.

The runtime status was confirmed as **Running**.

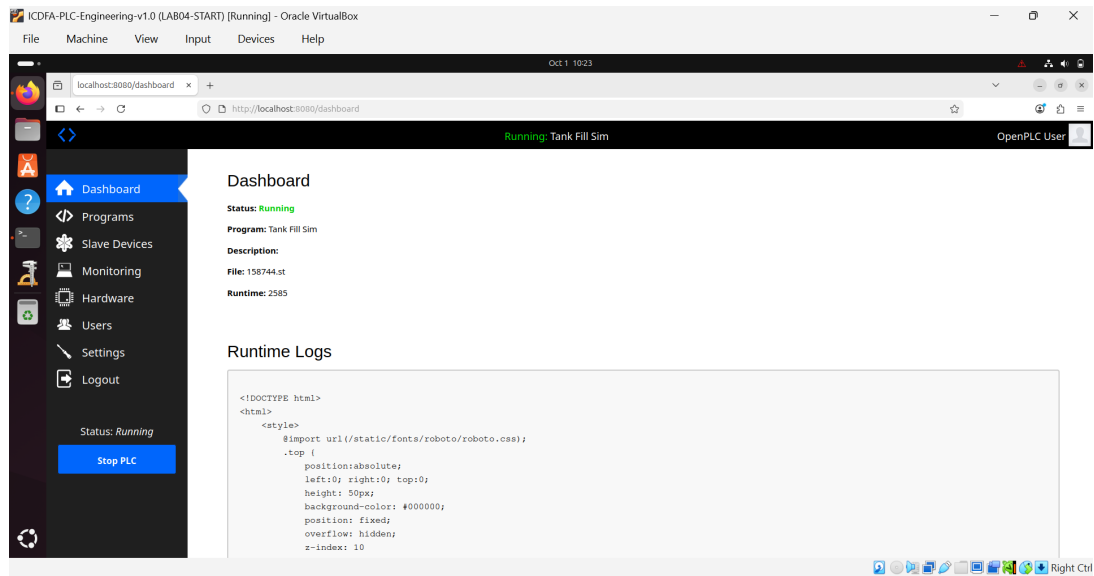


Figure 1: OpenPLC Runtime showing the Tank Fill Sim program in the Running state.

4 PLC–SCADA Communication

The PLC and SCADA virtual machines were connected through the isolated laboratory network. Connectivity between the two machines was verified.

The OpenPLC Runtime provided Modbus TCP communication on port 502. FUXA successfully detected the OpenPLC endpoint through the automatic discovery mechanism.

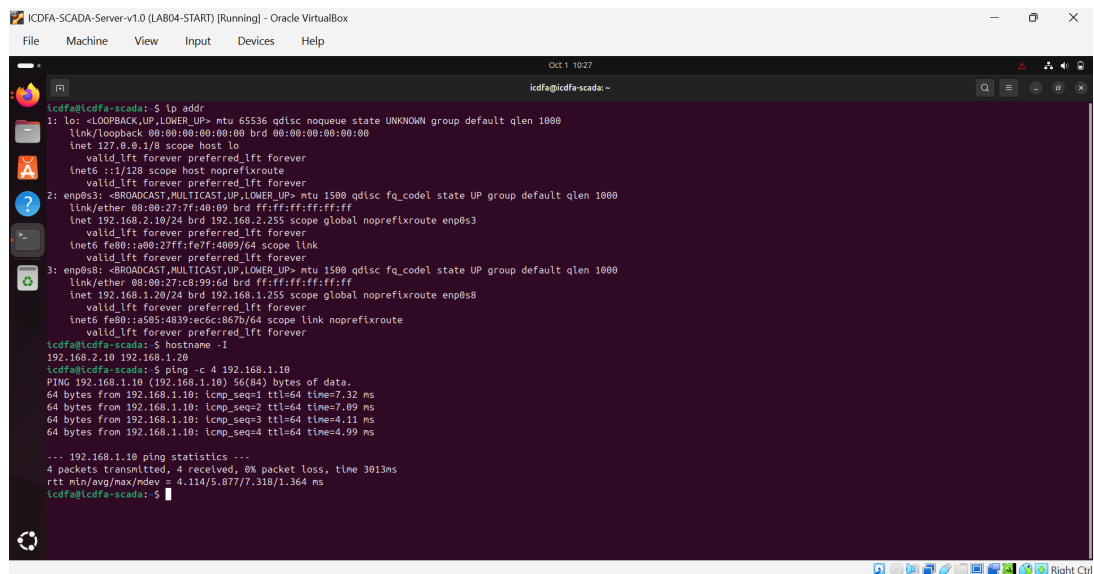


Figure 2: FUXA detecting the OpenPLC endpoint through automatic discovery.

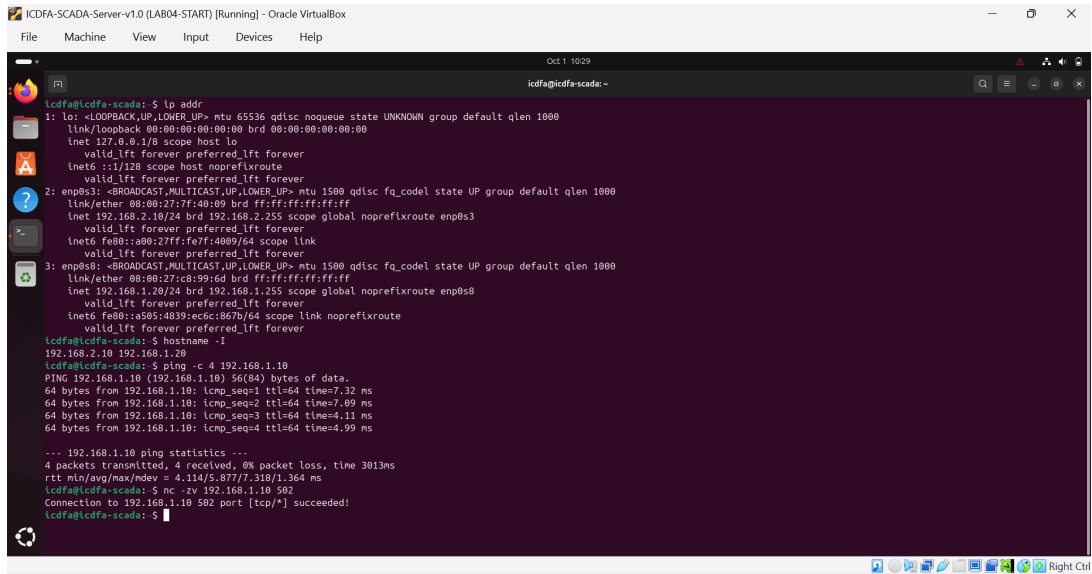


Figure 3: OpenPLC Modbus TCP communication and accepted client connection.

5 FUXA SCADA/HMI Operation

The FUXA project was started using the Play function. The HMI displayed the tank process, valve indication and other operator-facing information.

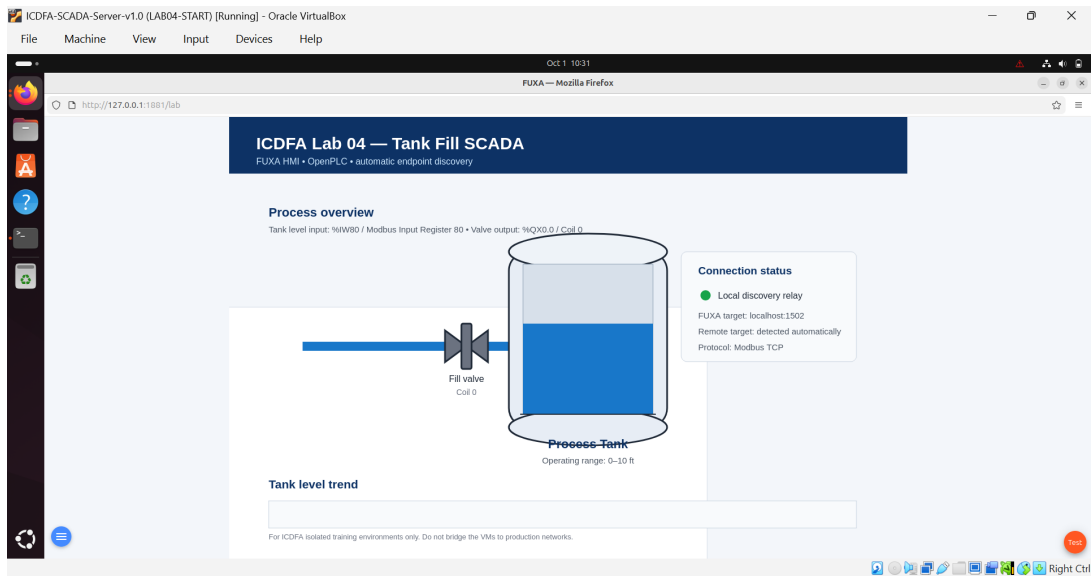


Figure 4: FUXA SCADA/HMI displaying the tank-fill process.

The communication path between the PLC and SCADA was successfully established. The HMI process screen was displayed and the OpenPLC endpoint remained connected. During the final observation, the expected changing tank-level value was not clearly reflected in the HMI. This indicates a process-value or register-mapping issue rather than

a failure of the PLC–SCADA network communication.

6 SCADA Architecture

The architecture of the laboratory can be represented as follows:

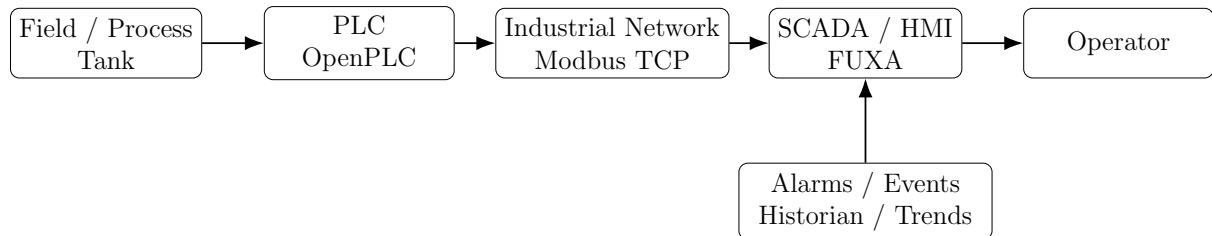


Figure 5: Simplified SCADA architecture.

7 Knowledge Check

1. What is the primary purpose of an HMI in a SCADA system?

The primary purpose of an HMI is to provide a graphical interface for monitoring and interacting with an industrial process. It presents process information in a form that can be easily understood by the operator.

2. What is the difference between an alarm, an event record and historized process data?

An alarm indicates that a process condition requires attention. An event record documents an occurrence or state change in the system. Historized process data stores process values over time so that previous conditions and trends can be reviewed.

3. Why should a SCADA operator have context rather than only raw sensor values?

Context helps the operator understand the meaning and significance of a sensor value. Information such as equipment status, alarms and trends helps determine whether a process condition is normal or requires attention.

4. What role does the PLC play compared with the SCADA application?

The PLC executes the control logic and controls the industrial process. The SCADA application supervises the process by displaying information to the operator and providing functions such as visualization, alarms, events and trends.

8 PLC Control Logic vs. HMI/SCADA Visualization

The PLC is responsible for executing the programmed control logic and controlling the industrial process. In this laboratory, OpenPLC runs the Tank Fill Sim program and manages the process and valve states.

The SCADA/HMI layer performs supervisory functions. FUXA communicates with the PLC through Modbus TCP and presents process information to the operator using graphical elements such as the tank-level indicator and valve status.

Therefore, the PLC performs the control and automation tasks, while the SCADA/HMI provides visualization and operator context. SCADA can also provide alarms, event recording and historical trends. This separation allows the PLC to execute control logic while the SCADA system provides monitoring and supervisory information to the operator.

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10 Conclusion

The laboratory demonstrated the basic relationship between a PLC, industrial communication network and SCADA/HMI.

The OpenPLC Runtime was successfully started, and FUXA successfully detected the OpenPLC endpoint through the automatic discovery mechanism. The Modbus TCP communication path was established and the FUXA HMI was loaded to provide supervisory visualization of the tank-fill process.

The laboratory also demonstrated the difference between PLC control logic and SCADA supervisory visualization.